

- Portfolio algorithms (*Continued*)
 - residual schedule exponential, 309–310
 - second wave of, 336
 - techniques, 306–316
 - third wave of, 337–338
 - trade schedule exponential, 308–309
 - trading rate parameter, 310–312
 - transaction costs in, 335–338
- optimization formulation, 302–306
 - constraint description, 303–306
 - objective function difficulty, 305
 - optimization objective function simplification, 305–306
- residual risk curve, 320–321
- trader's dilemma, 298–299
- transaction cost equations, 300–302
 - market impact, 301
 - one-sided optimization problem, 302
 - price appreciation, 301
 - timing risk, 302
- Portfolio algorithms, 297. *See also* Basket algorithms
- Portfolio attribution phase, 15
- Portfolio construction, 331
 - best execution frontier, 351–352
 - cost-adjusted frontier, 348–350
 - determining appropriate level of risk aversion, 350–351
- phase, 15
- portfolio management process, 339–341
- portfolio optimization and constraints, 332–335
- trading decision process, 341–343
- transaction costs in portfolio optimization, 335–338
- unifying investment and trading theories, 343–348
- with transaction costs, 352–359
 - expected stock return, 355
 - risk of a portfolio, 355–359
- Portfolio managers, 15, 25, 27–28, 50, 93, 193–194, 235, 262, 280, 298, 332–333, 337–338, 343, 358–359, 361, 363–366, 373, 378, 384–386, 389, 393*r*, 395–398, 415, 419–420
- Portfolio trading desk. *See* Program trading desk
- Post-trade analysis, 38, 96
 - adaptation tactic, 114–115
 - benchmark price performance, 106
 - cost versus profit and loss, 97
 - index adjusted performance metric, 111–112
 - market cost adjusted z-score, 113–114
 - measuring/forecasting, 96–97
 - participation weighted price (PWP) benchmark, 108–109
 - performance evaluation, 105–115
 - pre-trade benchmark, 110–111
 - relative performance measure (RPM), 109–110
 - trading price performance, 105–106
 - VWAP benchmark, 106–108
 - z-score evaluation metric, 112–113
- Pre-trade analysis, 37, 95–96
- Pre-trade benchmark, 110–111
- Pre-trade cost calculation, 365
- Pre-trade impact models, 365
- Pre-trade modeling approach, 367–374
 - applications, 372–374
 - cost estimation process, 368–371
- Pre-trade of pre-trades, 156, 367–374
- Price appreciation, 91
- Price change, 194
- Price impact, 166–167
- Price improvement, 279*f*, 281–283
- Price returns, 194, 196, 203–204
 - data sample, 203–204
 - standard deviation of, 196
- Price trajectory, 131–132
- Price volatility, 193–194, 240
 - ARCH volatility model, 205–206
 - definitions, 194–198
 - average return, 194–196
 - beta, 198
 - correlation, 197
 - covariance, 196
 - dispersion, 197
 - implied volatility, 198
 - log price return, 194
 - percentage price return, 194
 - price returns/price change, 194
 - value-at-risk, 197
 - volatility, 196
 - estimation results, 209
- exponential weighted moving average (EWMA), 205
- factor models, 221–226
 - constructing factor independence, 224–225
 - covariance estimation using, 225–226
 - cross-sectional multi-factor models, 229–231
 - index models, 227–228
 - macroeconomic factor models, 228–231
 - matrix notation, 223–224
 - multi-index factor model, 228
 - single index model, 227–228
 - statistical factor models, 231–234
 - types of, 227–234
- GARCH volatility model, 206
- historical moving average (HMA), 204–205
- HMA-VIX volatility forecasting model, 206–209
- likelihood function, 208–209
- market observations, 199–202
- measuring model performance, 209–221
- outlier analysis, 211
- price returns, 203–204
 - data sample, 203–204
- reliability on historical market data, problems from, 214–221
 - degrees of freedom, 219–221
 - false relationships, 214–218
- results, 211–214
- root mean square error (RMSE), 210
- root mean z-score squared error (RMZSE), 210
- stock volatility, forecasting, 202–206
 - volatility models, 202–206
- Price-based adaptation tactic, 32
- Price-based algorithm, 283
- Price-size models, 57
- Price-time models, 57
- Pricing models, 56–57
 - commission-based model, 57
 - maker-taker model, 56
 - taker-maker model, 56
- Principal bid, 24
- Principal component analysis (PCA), 27, 231–232
- Problem, researching, 164, 166–171
- “Profit and loss” algorithm, 2, 97
- Program sales trader, 27–28
- Program trading, 10–11, 25–26, 84
- Program trading desk, 25–26
- Proper categorization, 127
- Proper statistical tests, 126